1. What is inferential statistics?

Inferential statistics uses data from a sample to draw conclusions or make estimates about a larger population. It uses methods such as confidence intervals and hypothesis tests while accounting for sampling uncertainty.

2. What is hypothesis testing and its components?

Hypothesis testing is a statistical procedure used to evaluate a claim about a population using sample data. Its main components are the null hypothesis (H0), alternative hypothesis (H1), significance level (alpha), test statistic, p-value or critical value, and the final decision.

3. Explain confidence interval and critical value.

A confidence interval is a range of plausible values for a population parameter, calculated from sample data at a stated confidence level. A critical value is a cutoff from a probability distribution used to decide whether a test statistic falls in the rejection region.

4. Define p-value.

A p-value is the probability, assuming the null hypothesis is true, of obtaining a result at least as extreme as the observed result. A small p-value provides evidence against the null hypothesis; it is not the probability that the null hypothesis is true.

5. Differentiate Type I and Type II errors.

A Type I error occurs when a true null hypothesis is rejected (a false positive); its probability is denoted by alpha. A Type II error occurs when a false null hypothesis is not rejected (a false negative); its probability is denoted by beta.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

A z-test is commonly used to test means or proportions when its assumptions are met and the population standard deviation is known or a suitable large-sample approximation applies. A t-test compares means when the population standard deviation is unknown. A chi-square test examines categorical frequencies or independence between categorical variables. ANOVA compares the means of three or more groups.

7. What is Covariance?

Covariance measures how two variables vary together. A positive covariance means they tend to move in the same direction, while a negative covariance means they tend to move in opposite directions; its magnitude depends on the variables' scales.

8. What is Correlation?

Correlation measures the strength and direction of association between two variables. Pearson's correlation coefficient ranges from -1 to +1, with values near zero indicating little or no linear association; correlation alone does not prove causation.